



Notes Management System



Project Overview

The **Notes Management System** is a full-stack web application developed using **Flask (Python)** and **MySQL**.

It allows users to securely create, manage, and organize their personal notes with full CRUD functionality.

Each user has a private workspace, ensuring data security and isolation.



Abstract

This application enables users to:

- Register and log in securely
- Create, view, update, and delete notes
- Manage personal notes in a private environment

It includes authentication, session management, password encryption, and database integration, making it a complete and modern web solution.



Objectives

- Build a user-friendly notes management platform
 - Implement secure authentication using Flask sessions
 - Perform CRUD operations on notes
 - Ensure user-specific data privacy
 - Integrate Flask with MySQL database
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Tech Stack

Layer	Technology	Purpose
Frontend	HTML5, CSS3, Bootstrap 5	UI Design & Responsiveness
Backend	Flask (Python)	Application Logic & Routing
Database	MySQL	Data Storage
Security	Werkzeug	Password Hashing
Server	Flask Development Server	Local Development & Testing

Installation & Setup

1 Clone the Repository

```
git clone https://github.com/your-username/notes-management-system.git
cd notes-management-system
```

2 Create Virtual Environment

```
python -m venv venv
venv\Scripts\activate    # Windows
```

3 Install Dependencies

```
pip install -r requirements.txt
```

4 Run the Application

```
python app.py
```

Requirements

```
blinker==1.9.0
click==8.4.1
colorama==0.4.6
Flask==3.1.3
Flask-Mail==0.10.0
itsdangerous==2.2.0
Jinja2==3.1.6
MarkupSafe==3.0.3
mysql-connector-python==9.7.0
PyMySQL==1.2.0
Werkzeug==3.1.8
```

Project Modules

User Module

- Register new account
- Login with credentials
- Logout securely
- Session management using Flask

Notes Module (CRUD)

Operation	Route	Description
Create	/addnote	Add a new note
Read All	/viewall	View all user notes
Read One	/viewnotes/	View single note
Update	/updatenote/	Edit existing note
Delete	/deletenote/	Delete note permanently

Database Structure

Users Table

Column	Type	Description
id	INT (PK)	Unique user ID
username	VARCHAR(100)	Unique username
email	VARCHAR(100)	User email
password	VARCHAR(255)	Encrypted password

Notes Table

Column	Type	Description
id	INT (PK)	Unique note ID
title	VARCHAR(100)	Note title
content	TEXT	Note content

Column	Type	Description
created_at	TIMESTAMP	Creation time
user_id	INT (FK)	Linked to users.id



Security Features

- Passwords are hashed using `generate_password_hash()`
- User-specific note access
- Session-based authentication
- Protected routes (login required)
- ON DELETE CASCADE for database integrity



Frontend Features

- Responsive UI using Bootstrap 5
- Clean navigation bar and footer
- Flash messages for user feedback
- Simple and beginner-friendly design



Application Workflow

1 User Registration

User signs up → Data stored securely in database

2 Login

Credentials verified → Session created

3 Notes Management

User performs CRUD operations on personal notes

4 Logout

Session cleared → Redirect to login page



Learning Outcomes

- Build full-stack apps with Flask & MySQL
- Understand CRUD operations
- Use Jinja2 templating
- Implement authentication & sessions

- Work with relational databases
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Conclusion

This project demonstrates how to build a secure, database-driven web application using Flask. It is ideal for beginners learning **Full Stack Python Development**.

Future Improvements

- Add search functionality
 - Add categories/tags for notes
 - Deploy on cloud (AWS/Render)
 - Add REST API support
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Author

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License

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